RM – Credit Instruments and Risk Homework

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1 Exercise 1

Consider a corporate bond:

- with notional N (compare to \$1 in the lecture notes);
- issued by a firm with constant default intensity λ (not a function of time);
- final maturity T, at which the notional is returned to the bond holders;
- constant continuous coupon c, i.e. in any time interval $[t, t + dt]$, where $t < T$, the bond holders are paid $N \cdot c \cdot dt$ as long as there is no default prior to t;
- recovery rate R, i.e. at the time of default (if before T), the bond holders receive $N \cdot R$.

Assume the risk-free rate is constant at $r > 0$.

$$\begin{aligned} \mathbb{P}(\tau > t) &= e^{-\lambda t}, \\ \mathbb{P}(\tau \in dt) &= \lambda e^{-\lambda t} dt \end{aligned}$$

Case 1: $R = c = 0$

Question: What is the bond value at $t = 0$?

Solution:

$$\begin{aligned} V_0 &= N e^{-rT} \mathbb{P}(\tau > T) \\ &= N e^{-rT} e^{-\lambda T} \\ &= N e^{-(r+\lambda)T} \end{aligned}$$

$$\boxed{V_0 = N e^{-(r+\lambda)T}}$$

Case 2: $R = 0,\ c \neq 0$

Question:

(a) What is the value of the coupon stream at $t = 0$?

(b) What is the total value of the bond at $t = 0$?

(c) What is the limit of the initial bond value as $T \to \infty$?

(d) At which value of c is the initial bond value equal to N (par coupon)?

Solution:
(a) Coupon value

$$\begin{aligned} V_{\text{coupon}} &= \int_0^T Nc\, e^{-rt}\mathbb{P}(\tau > t)\, dt \\ &= Nc \int_0^T e^{-rt} e^{-\lambda t} dt \\ &= Nc \int_0^T e^{-(r+\lambda)t} dt \\ &= \frac{Nc}{r+\lambda}\left(1 - e^{-(r+\lambda)T}\right) \end{aligned}$$

(b) Total value

$$\begin{aligned} V_0 &= V_{\text{coupon}} + V_{\text{notional}} \\ &= \frac{Nc}{r+\lambda}\left(1 - e^{-(r+\lambda)T}\right) + Ne^{-(r+\lambda)T} \end{aligned}$$

$$\boxed{V_0 = \frac{Nc}{r+\lambda}\left(1 - e^{-(r+\lambda)T}\right) + Ne^{-(r+\lambda)T}}$$

(c) Limit as $T \to \infty$

$$\boxed{\lim_{T\to\infty} V_0 = \frac{Nc}{r+\lambda}}$$

(d) Par coupon

$$1 = \frac{c}{r+\lambda}\left(1 - e^{-(r+\lambda)T}\right) + e^{-(r+\lambda)T}$$

$$\boxed{c = r + \lambda}$$

Case 3: $R \neq 0,\ c \neq 0$

Question:

(a) What is the value of the recovery component at $t = 0$?

(b) What is the total value of the bond at $t = 0$?

(c) What is the limit of the initial bond value as $T \to \infty$?

(d) At which value of c is the initial bond value equal to N (par coupon)?

Solution:
(a) Recovery value

$$\begin{aligned} V_{\text{recovery}} &= \int_0^T NR\, e^{-rt} \lambda e^{-\lambda t} dt \\ &= NR\lambda \int_0^T e^{-(r+\lambda)t} dt \\ &= \frac{NR\lambda}{r+\lambda}\left(1 - e^{-(r+\lambda)T}\right) \end{aligned}$$

(b) Total value

$$\begin{aligned} V_0 &= V_{\text{coupon}} + V_{\text{notional}} + V_{\text{recovery}} \\ &= \frac{N(c + R\lambda)}{r + \lambda}\left(1 - e^{-(r+\lambda)T}\right) + Ne^{-(r+\lambda)T} \end{aligned}$$

$$\boxed{V_0 = \frac{N(c + R\lambda)}{r + \lambda}\left(1 - e^{-(r+\lambda)T}\right) + Ne^{-(r+\lambda)T}}$$

(c) Limit as $T \to \infty$

$$\boxed{\lim_{T\to\infty} V_0 = \frac{N(c + R\lambda)}{r + \lambda}}$$

(d) Par coupon

$$1 = \frac{c + R\lambda}{r + \lambda}\left(1 - e^{-(r+\lambda)T}\right) + e^{-(r+\lambda)T}$$

$$\boxed{c = r + \lambda(1 - R)}$$